

- 56** Sickle cell anaemia is a recessive genetic condition that results in abnormally-shaped red blood cells due to the production of a faulty type of haemoglobin. Children born with sickle cell anaemia rarely live to adulthood without significant medical intervention. Carriers, who only have one copy of the sickle cell allele, have greater resistance to the disease malaria than people with two copies of the allele for normal functional haemoglobin.

Using this information, which of the following statements is/are correct?

- 1** People with sickle cell anaemia would have reduced anaerobic respiration in their muscle cells.
 - 2** In areas with malaria the percentage of people surviving with sickle cell anaemia increases.
 - 3** In parts of Africa where malaria is more common you would expect to find more people with a sickle cell allele.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3